CM SERDES GMSL2/3 OVERVIEW & MAIN FEATURES

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#OneStepAhead

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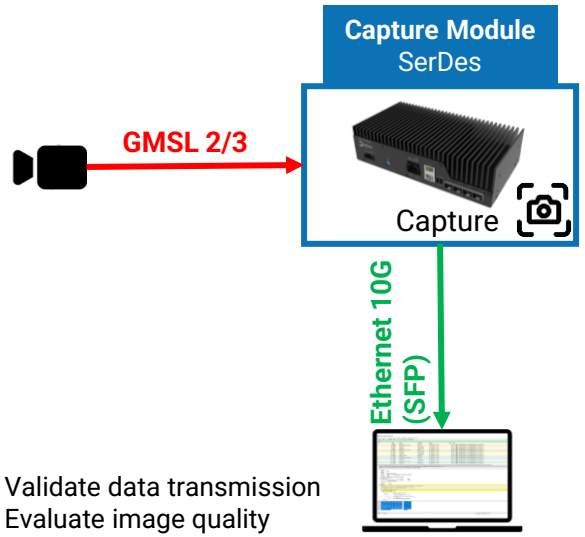
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CAPTURE MODULE SERDES

CAPTURE MODULE SERDES USE-CASES

Our CM line-up of products is designed to cover the 3 main use cases for **ADAS applications**

Sensor Module Testbenches



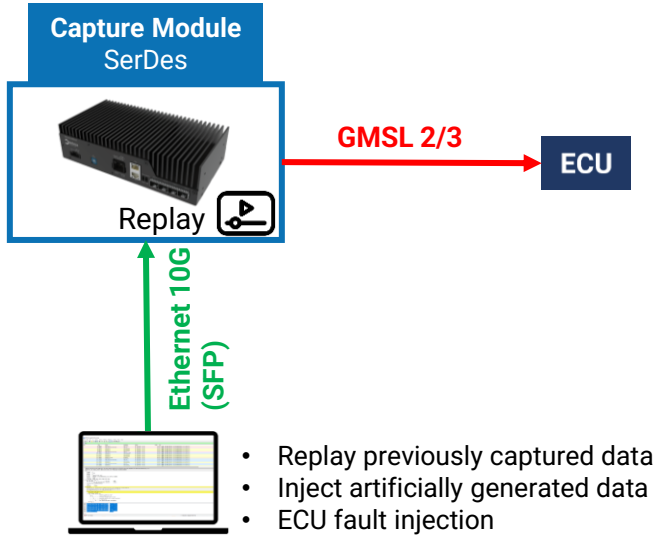
- Validate data transmission
- Evaluate image quality

Host Test Server

Targeted for:

- Sensor product validation plans
- Sensor image quality validation
- Sensor production test

ADAS ECU Testbenches



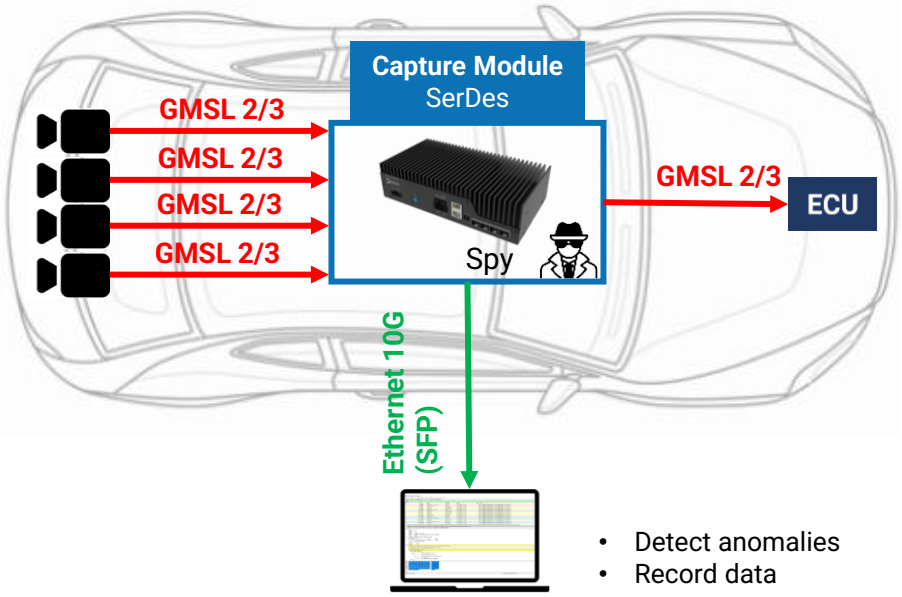
- Replay previously captured data
- Inject artificially generated data
- ECU fault injection

Host Test Server

Targeted for:

- AI perception algorithms validation
- ADAS ECU product validation plans
- ADAS ECU fault injection tests
- ADAS ECU production test

Prototype Vehicle



- Detect anomalies
- Record data

Data Logger

Targeted for:

- Vehicle integration testing
- Fleet car testing
- Data recording plan for AI training

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CAPTURE MODULE SERDES GMSL2/3

MAIN FEATURES





Main Features:

- 4x **GMSL2/3** link (8 ports)
- Easy configuration via **webserver**
- Network Time Synchronization (i.e. **gPTP**)
- **Lossless JPEG Compression**
- Hardware **Timestamping**
- **Wake-up** capable
- High voltage range: 12 to 24 volt DC
- **Galvanic Isolated** DC/DC
- Heatsink integrated on case
- Size: 240mm X 130mm X 55mm
- Quad mini coax connector for 4 camera GMSL2/3 links (**GMSL2/3** compatible Serializer)
- Quad mini coax connector for 4 ECU GMSL2/3 links (**GMSL2/3** compatible Deserializer)

Already existing product in use by major OEM during several years for its data recording plans

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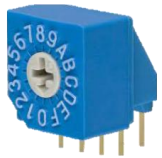
CONNECTORS

DataLogger Interfaces



Isolated Power Connector
- Isolated Power input
- 1x Wake signals (In)

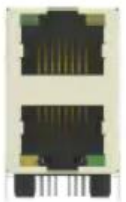
Rotary Switch
- Device IP Configuration



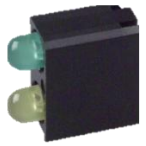
Main Connector
- Power
- 100Base-T1 Backup Config Port
- 2x Wake signals (In/Out)
- Tyco MQS 1-967250-1



2x Ethernet RJ45 (1Gbps)
- Configuration
- Synchronization



4x Status LEDs
- Microcontroller status
- FPGA Status
- Synchronization status
- 100Base-T1 Config Port link up



4x SFP+ Ports (10Gbps)
- Logging



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CAPTURE MODULE SERDES GMSL2/3

CONNECTORS

Vehicle Interfaces



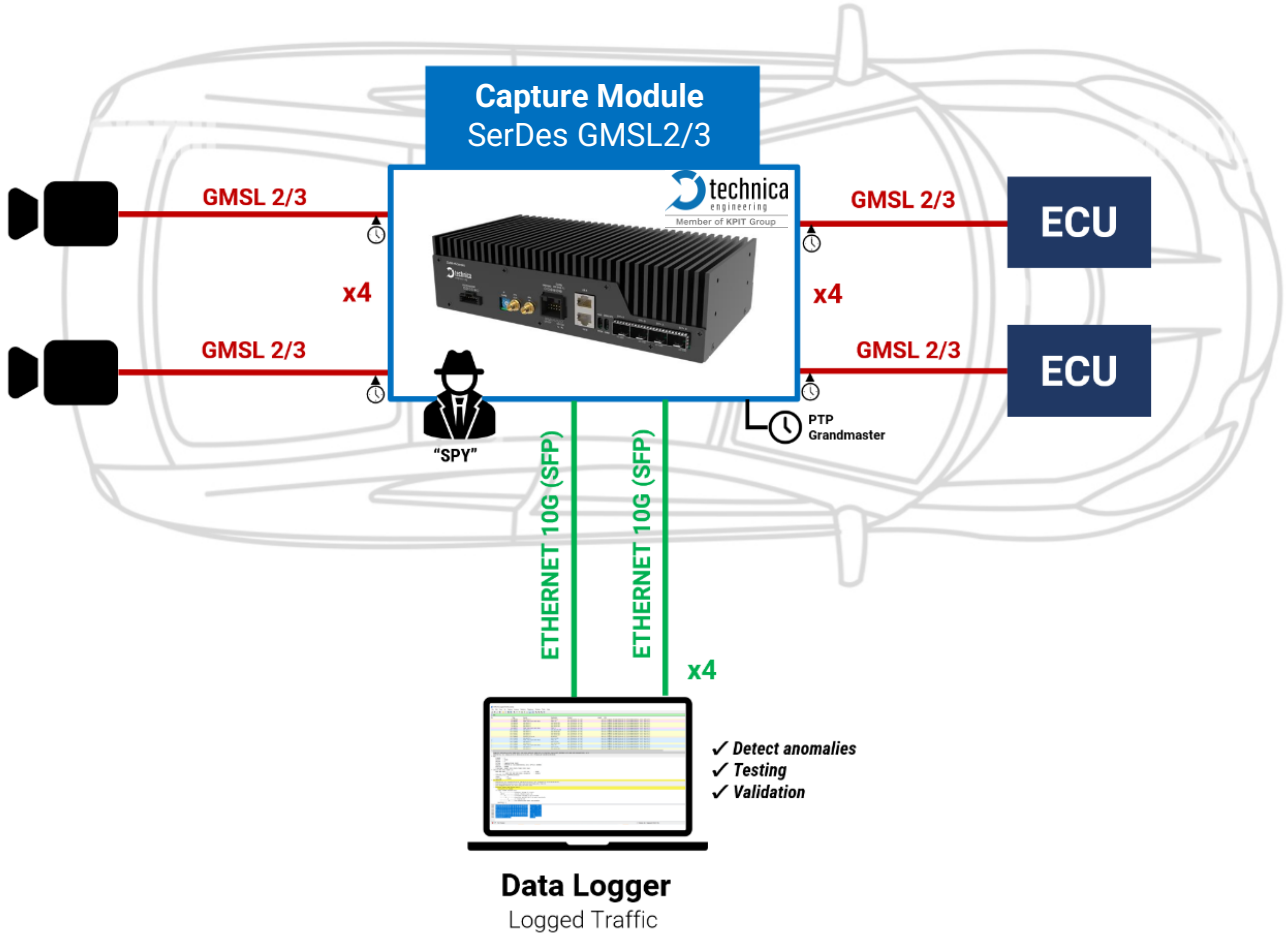
GMSL2/3 connector

- Quad mini coax connector for 4 camera GMSL2/3 links (**GMSL2/3** compatible Serializer)
- Quad mini coax connector for 4 ECU GMSL2/3 links (**GMSL2/3** compatible Deserializer)
- Tyco **MATE-AX** connector
- Camera **POC** options:
  - Provided from ECU (bypassed through SERDES CM)
  - Injected by SERDES CM (Adjustable POC voltage)

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CAPTURE MODULE SERDES GMSL2/3

MAIN USE CASE



Overview

Spy on GMSL2/3 traffic from various links in the car

- ✓ Allow the traffic to pass through the device with very low latency
- ✓ Send captured traffic encapsulated in Ethernet frames over 10G SFP+ interfaces to another system.

(*) GMSL2/3 is mainly used to transport video streams between camera sensors and ECUs.

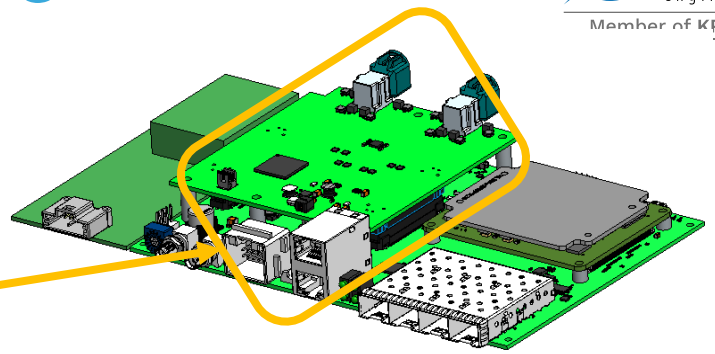
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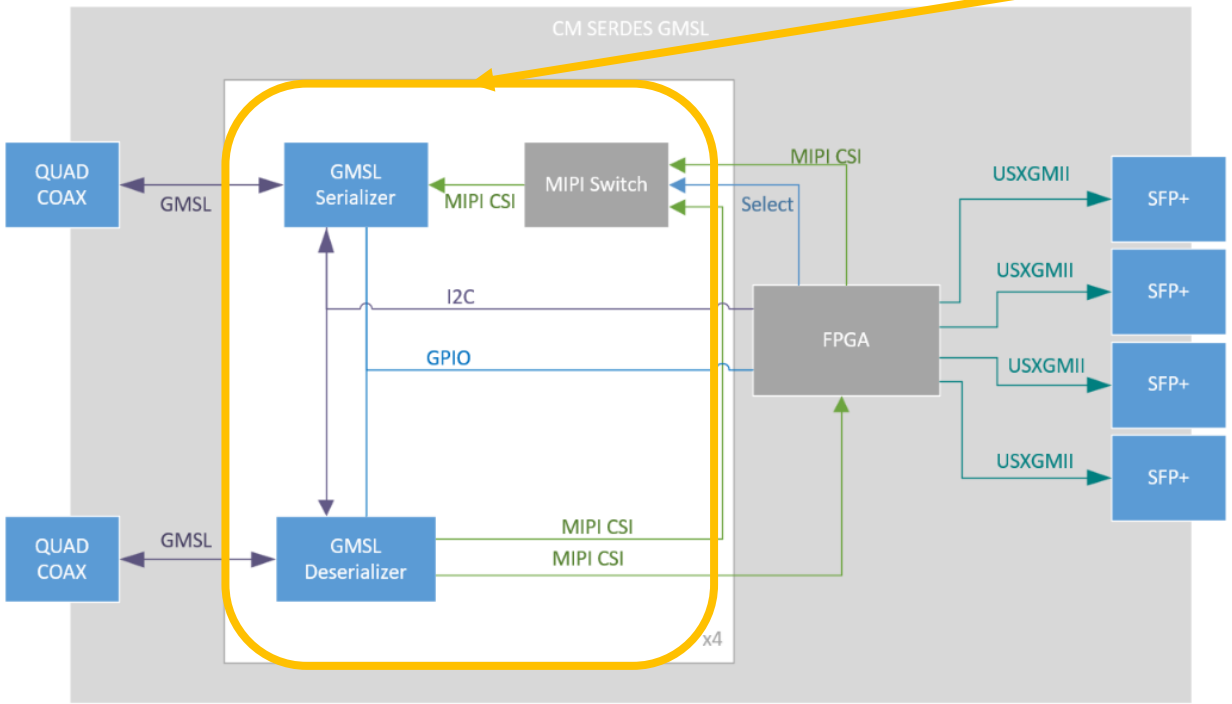
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CAPTURE MODULE SERDES GMSL2/3

SIMPLIFIED ARCHITECTURE

Modular design simplifying the creation of product variant for different physical interfaces





Versatile product designed to cover multiple use cases with the same device:

- **Sensor Module** Testbenches
- **ECU HIL** Testbenches (video Injection)
- **Prototype Vehicle** Tests (recording + AI)

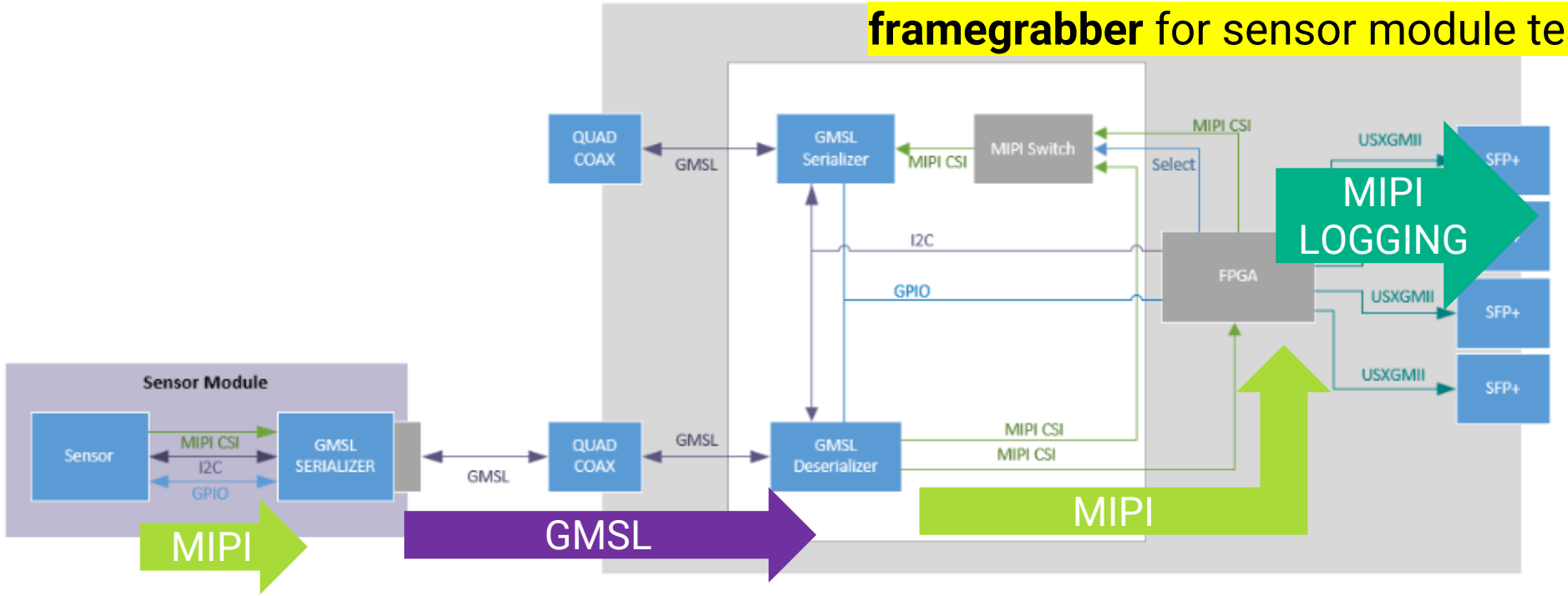
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CAPTURE MODULE SERDES GMSL2/3

USE CASE #1: **SENSOR MODULE** TESTBENCHES

Capture Module Serdes can function as a **framegrabber** for sensor module testing activities



Capture Module Serdes can be used for testing sensor modules without requiring a dedicated ECU, functioning as a **framegrabber**

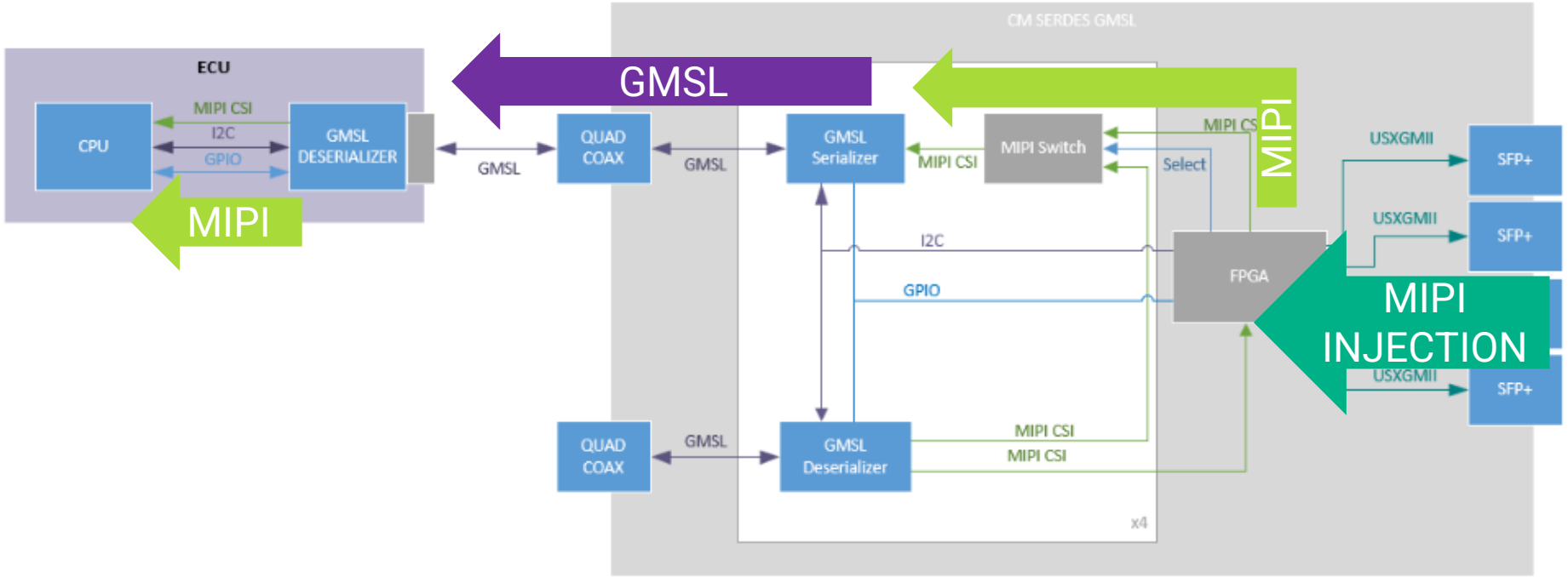
The CM Serdes can configuring the sensor module through a **Remote API** which can trigger I2C operations on demand, then **capturing the MIPI data** transmitted over GMSL and finally **packetizing it into ethernet frames** for logging.

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CAPTURE MODULE SERDES GMSL2/3

USE CASE #2: **ECU HIL** TESTBENCHES



Capture Module Serdes can be used on HILs for testing data processing ECU, **injecting data streams** to the ECU SerDes inputs **without requiring dummy sensors** on the HIL setups.

With the objective to verify how the ECU data processing applications behaves on a real hardware, the **CM Serdes can inject data**, from previously captured recordings, to the ECU SerDes inputs **simulating real-world scenarios on a HIL test bench**.

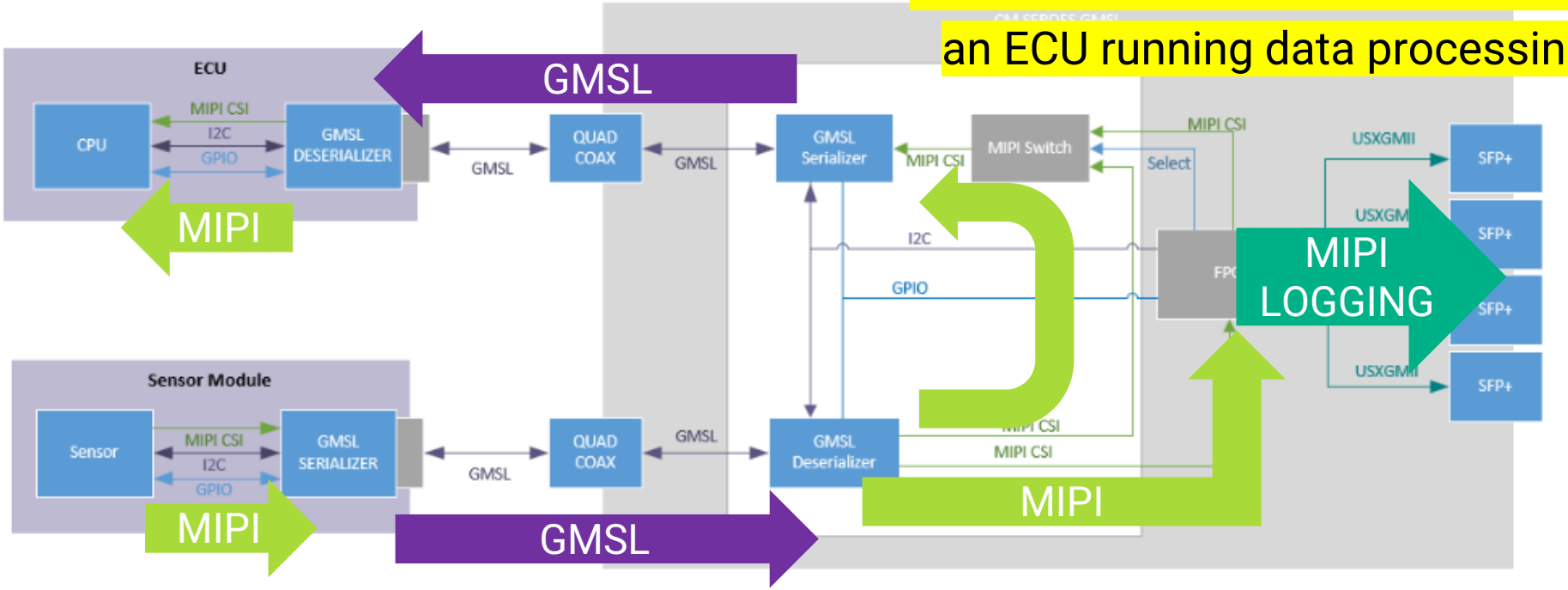
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CAPTURE MODULE SERDES GMSL2/3

USE CASE #3: **PROTOTYPE VEHICLE** INTEGRATION TESTS

Data stream can be recorded without interfering an ECU running data processing applications



Capture Module Serdes can be used on **Prototype Vehicle** integration tests to capture data transmitted between a sensor module and an ECU over the SerDes interface while the data is forwarded without modification and minimal latency.

This operating mode allows the ECU to run its application software, processing the data transmitting from the sensors, while the data is being recorded for later analysis.

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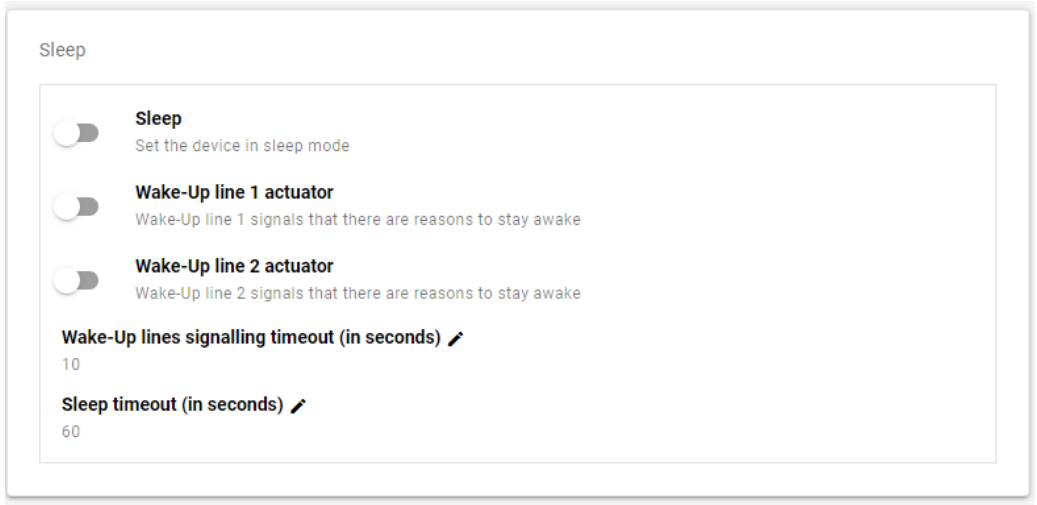
CAPTURE MODULE SERDES GMSL2/3

BASIC FEATURES: COMMON CM FEATURES

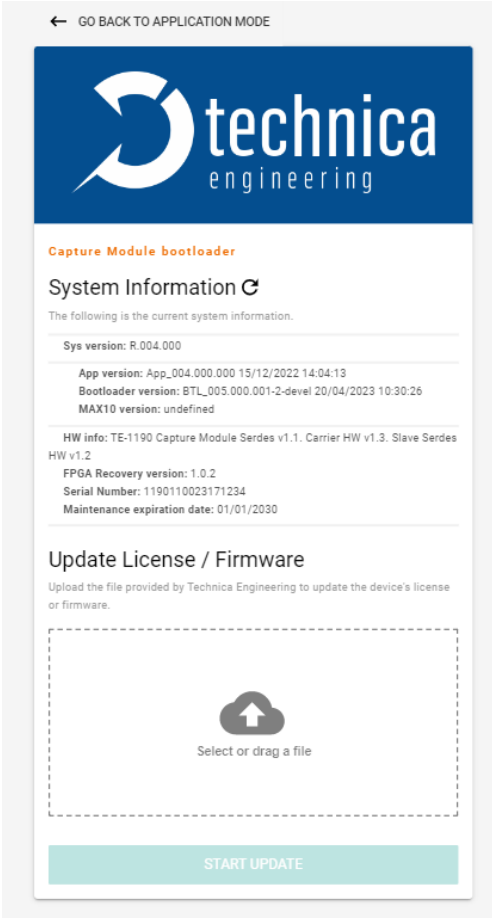
Synchronization through GB ports:



Sleep mode, including keep awake reason for all channels



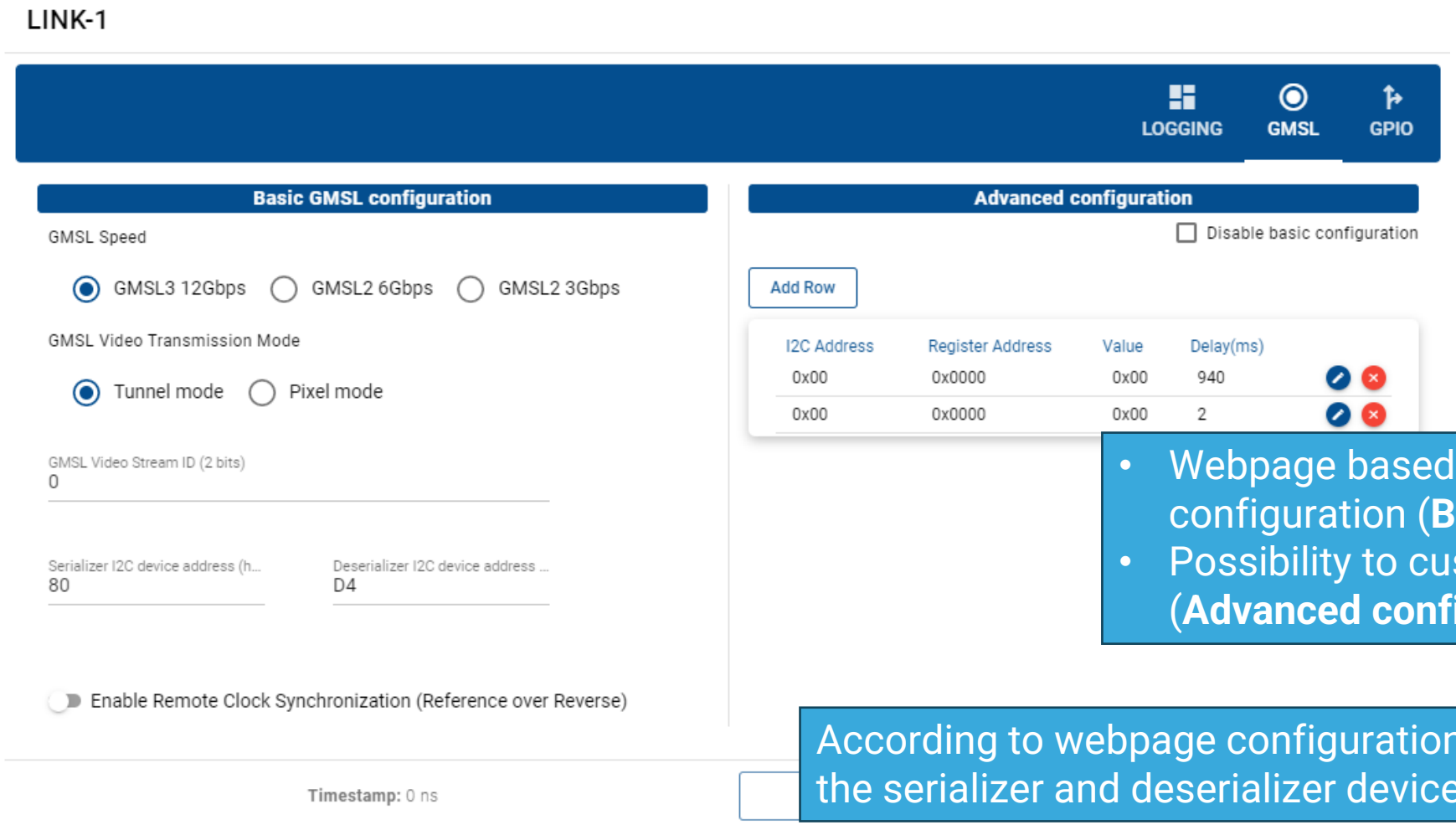
Webpage based **bootloader** for easy software update.



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CAPTURE MODULE SERDES GMSL2/3

BASIC FEATURES: LINK CONFIGURATION



- Webpage based wizard for Serdes IC configuration (**Basic configuration**)
- Possibility to customize initial configuration (**Advanced configuration**)

According to webpage configuration, the uC (calypso) initializes the serializer and deserializer devices on the CM at every power

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CAPTURE MODULE SERDES GMSL2/3

BASIC FEATURES: SUPPORT FOR VC & METADATA

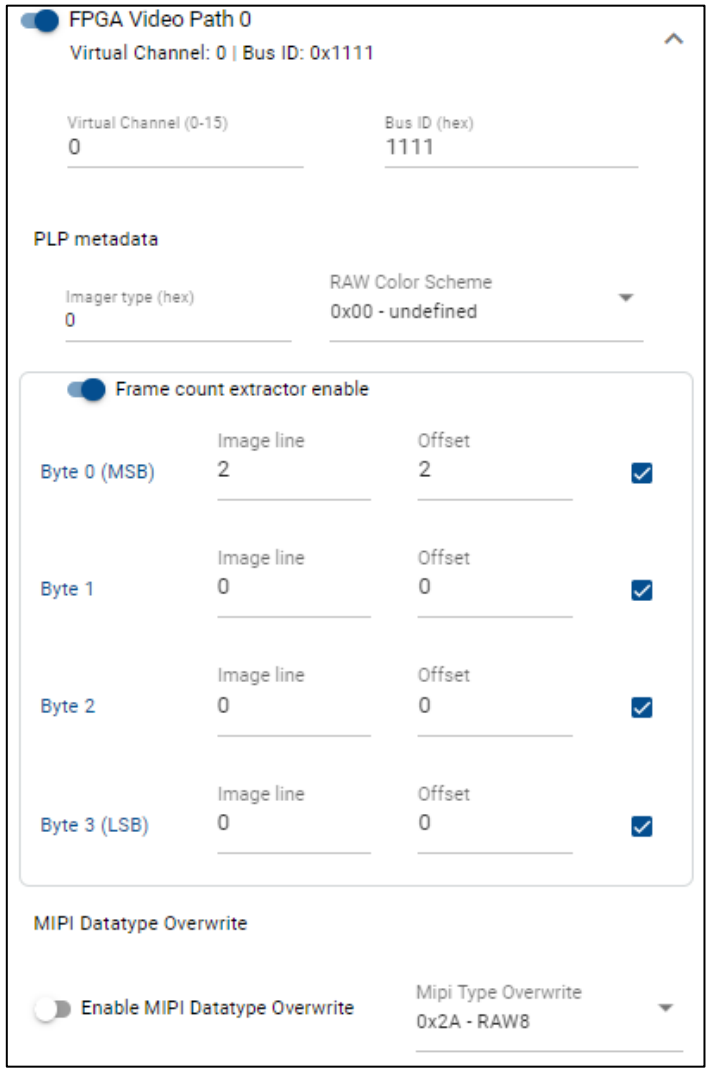
Logging Support for 4 **Virtual Channels** according to MIPI specification on the same link

- Independent BusId for each Virtual Channel
- Independent Metadata configuration for each Virtual Channel

Generation of ethernet **metadata packet** for each captured frame:

- **Resolution** (image width and height) computed from received MIPI packets
- **Pixel format** (according to MIPI Data Type)
- User defined information:
  - Imager Type
  - RAW Color scheme
  - Frame counter, extracted from embedded data MIPI packets

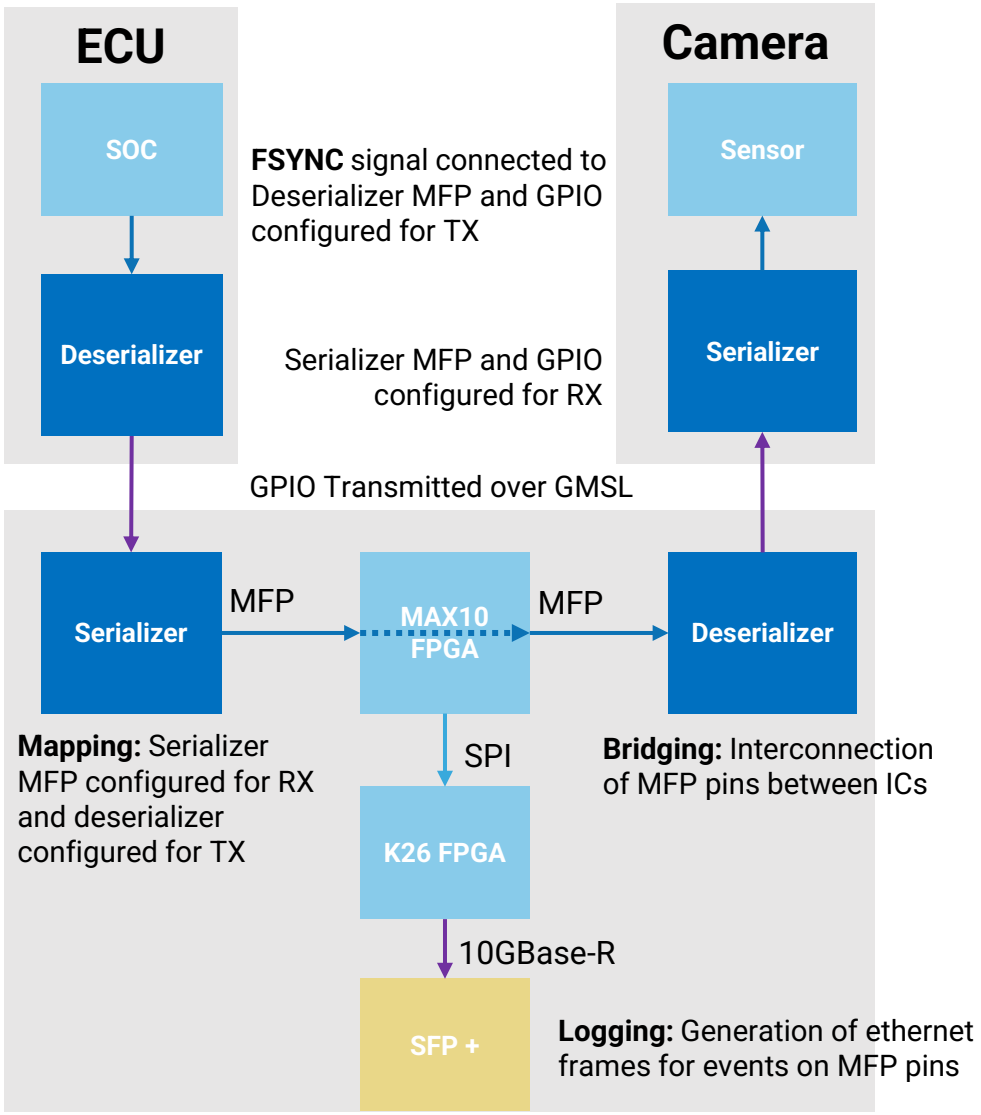
```
*************  Metadata  *************
Pix Format:   0x1e   Width:     2880
RAW Col S:    0x00   Height:    1859
Compr.:       0x00   Im Type:   0x0000000000000000
Err Flags: 0x0000   Fr Count:  0x00000000
```



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CAPTURE MODULE SERDES GMSL2/3

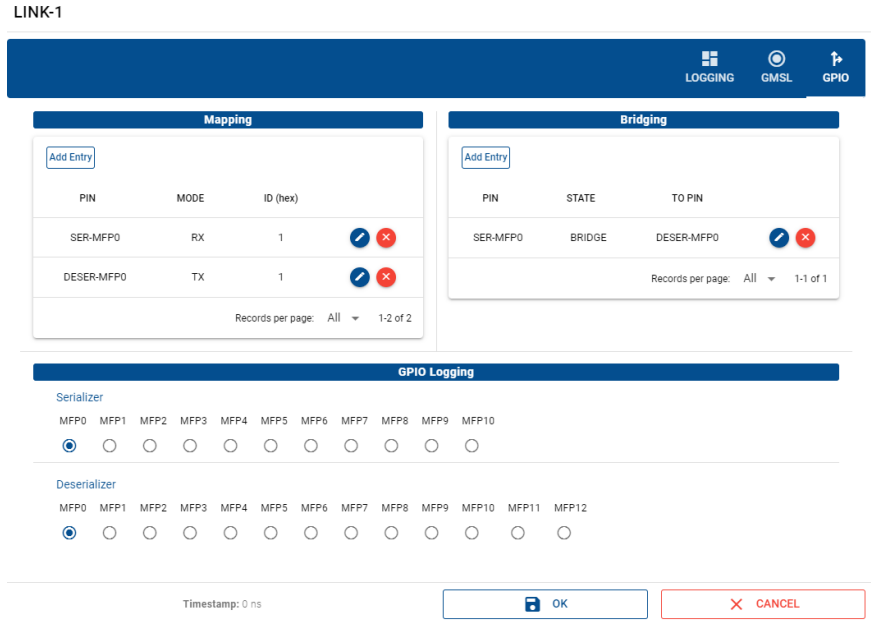
ADVANCED FEATURES: GPIO CONFIGURATION



GPIO main features:

- **Mapping**: Configuration of the serdes devices MFP pins for transmission or reception
- **Bridging**: Connection of MFP pins within the CM Serdes GMSL2/3
- **Logging**: capture events on enables GPIOs and log them through Ethernet frames

Configuration possible through webpage:



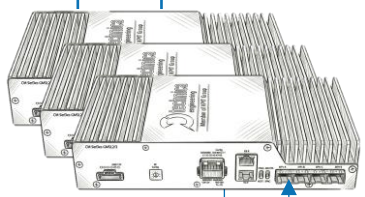
STATE OF THE ART PRODUCTS (TECHNICA IP)

FULL SETUP CM SERDES DEMO: FROM LOGGING VIDEO IN FLEETS TO HARDWARE IN THE LOOP (HIL) SYSTEMS

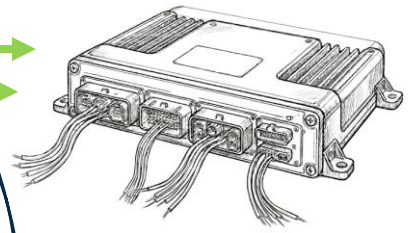
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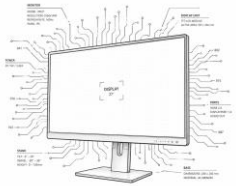


NVIDIA ORIN





PENDING



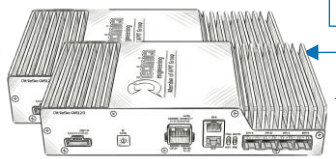
DISPLAY ADAS ECU PROCESSED VIDEO

CM SERDES LOGGING x3

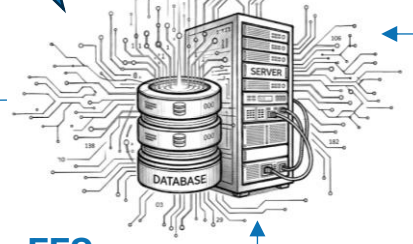
HIL

IPC

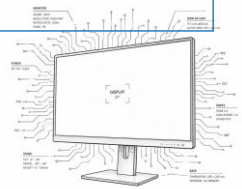




CM SERDES INJECTION GMSL2 x2



EES

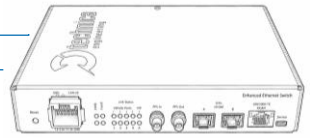


DISPLAY LOGGED VIDEO



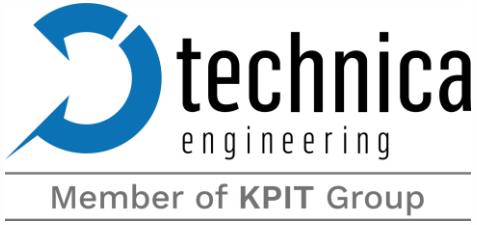


SYNC gPTP



SYNC gPTP

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